

Modeling of Electric Vehicle Loads Applicable to Power System Quasi-Steady State Analysis

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Abstract—Electric vehicles (EVs) are new power loads of power systems. It is important to model the charging process of EVs in power system numerical simulation. This paper proposes a methodology for modeling of EV loads applicable to power flow calculation and other quasi-steady state analysis. The numerical methodology of EV load modeling is expounded from the DC and the AC side of charging EVs. In the DC side, the constant current and the constant voltage charging processes are considered. In the AC side, the changes of the charging efficiency and power factor versus the AC frequency and voltage are analyzed. Based on the current EV market status and the development trends, four types of EV load models are calculated using this methodology. The EV load modeling methodology considers EV quantities, initial state-of-charge distributions, load change processes during charging and the effect of grid side parameters.

Keywords—power system; electric vehicle; numerical simulation; power flow calculation; model

I. INTRODUCTION

Power system numerical simulation is the main basis for power system operation, planning, and control. A valid simulation process can be used to improve the performance of power systems. Reliable numerical simulation results depend on proper system component models and stable numerical algorithms. Recently the quantity of electric vehicles (EVs) is increasing greatly because of their environmentally friendly nature. EVs are new power loads and it is important to study EVs' load model in steady state and transient state analysis, since it can improve the accuracy of power system numerical simulation and help to analyze the effect of plug-in EVs. There is little work on EV load modeling methodology that can be applied to power system numerical simulation. Based on these, this paper studies the EV load model applicable to power flow calculation, which is foundational for power system quasi-steady state analysis and further transient state analysis, when considering plug-in EVs in the grid.

EV load characteristics mainly include EV battery characteristics and charging circuit characteristics. The EV charging circuit establishes connections between the battery in the DC side and the EV load in the AC side through the charging efficiency and power factor. The lithium-ion battery model is used for studying due to its popularity among EV batteries. The DC side battery charging characteristics and the AC side electronic device charging characteristics will be analyzed separately, since the time-domain simulation for power electronic devices is time-consuming.

II. COMPONENTS OF ELECTRIC VEHICLE LOADS

Charging EVs are connected to the grid by battery chargers. A battery charger or an on-board battery management system (BMS) controls the charging process in order to keep the battery unharmed when charging with the scheduled current or voltage. Besides, battery aging also affects EV load characteristics. The brief introduction to components of electric vehicle loads is as follows.

A. Electric Vehicle Battery Chargers

There are two types of charging piles, AC charging piles and DC charging piles (also known as fast chargers). In China, the input to AC charging piles is single-phase 220 V AC power. The output regulated AC power (usually with AC 16 A) from AC charging piles flows into on-board battery chargers before charging batteries. The input to DC charging piles is three-phase 380 V AC power. The output is DC power (usually with DC 125 A or larger) and can be delivered to the battery directly. International charging level standards include 120 V/15 A, 240 V/40 A, and 480 V/90-250 A. The former two types use single-phase charging and the third one uses three-phase charging [1]. SAE Standard J1772 defines similarly EV charging levels [2]. This study concentrates on single-phase charging load model which is widely applied in residential side.

There are many studies on battery charger circuits. According to [3-6], on-board and fast chargers have similar circuit structure and components. The recommended basic charging circuit diagram of on-board chargers for simulation is presented in Fig. 1. The on-board charger circuit includes an AC filter, an active power factor correction (APFC) module and a buck converter. Hysteresis-band tracking technology can be used in both the APFC module and the buck converter stage to improve the power factor and keep constant current (CC) or constant voltage (CV) during charging. The isolation circuit is omitted in Fig. 1 for simplification, and it can be shown that this omission has little effect on the accuracy.

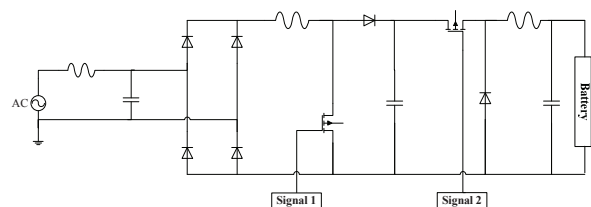


Fig. 1. Basic charging circuit diagram of on-board chargers for simulation

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B. Battery Characteristics of Electric Vehicles

The battery of EVs consists of many battery cells in series and parallel connections. Reference [7] describes the CC and CV charging processes, which are widely used in charging lithium-ion batteries [8]. The state of charge (SOC) of EV batteries is generally in 20% and above. In this study, CC charging is applied in 20%-90% SOC range and CV charging is applied in 90%-100% SOC range [9]. The charge and discharge characteristics of several types of batteries can be obtained by simulation using MATLAB SIMULINK software. The dynamic behavior of the lithium-ion battery model is validated by experiments and the simulated values and the real ones are perfectly matched when the SOC is in 20% to 100% [10]. Without loss of generality, in this study the default parameters of the lithium-ion battery in MATLAB are used.

The main characteristics of EV batteries are changes of the battery voltage against the SOC and the charge or discharge current. Take a lithium-ion battery with rated capacity (C) 92.8 Ah and nominal voltage 345.6 V as an example. Its voltage profiles in the CC charging period are shown in Fig. 2. The battery SOC satisfies the following equation

$$SOC = SOC_{initial} + \int I_c dt / C \quad (1)$$

where $SOC_{initial}$ is the initial SOC, I_c is the charge current (in A) and t is the charge time (in h). During the CV charging period, the charge current and the charge power decrease with time.

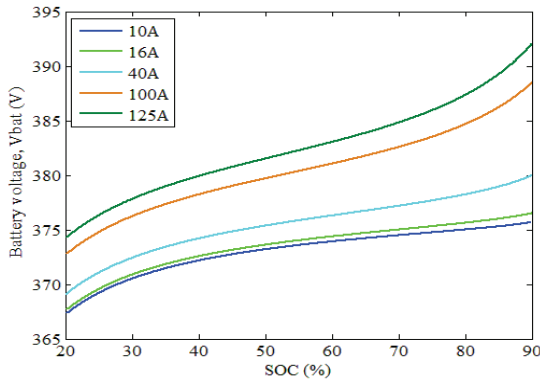


Fig. 2. Lithium-ion battery voltage profiles in the CC charge period

C. Battery Management System and Battery Aging

A battery management system (BMS) takes a significant role in EVs, whose primary functions include limiting over-charge and over-discharge, ensuring balance in cells, temperature control, and so on. The BMS could control battery charging process to accomplish smart charging by communicating with the charger. With the increase of battery operating cycle, the effect of battery aging on battery characteristics appears. The aging behavior of lithium-ion batteries mainly includes the decrease of capacity and the increase of internal resistance, which has a direct effect on battery power characteristics [11]. The battery models in MATLAB use constant capacity and constant internal resistance parameters. By referring to relevant researches, the battery model parameters can be revised to consider battery aging effect on load characteristics.

III. MODELING OF ELECTRIC VEHICLE LOADS IN THE DIRECT CURRENT SIDE

Two aspects need to be considered for the EV load model in power system quasi-steady state analysis. The first aspect is about the charging efficiency and power factor, which are related to the battery charger characteristics. The second aspect is the charging power demand mainly affected by the battery characteristics, which needs more consideration in EV load modeling since battery chargers are usually well-designed and their characteristics are similar while there are various types of batteries. In order to get the EV load model applicable to power flow calculation, it is expected to get the charging power of one single EV in the form of numerical formula. After that, by further considering the quantity and the initial SOC distribution of charging EVs, the expected EV load model will be obtained. In addition, the charging of EVs is centralized at some specific time period and it takes a certain time before EVs are fully charged. This regularity in EV loads makes it possible to estimate the change of EV loads in some period. Finally, with the development and subdivision of the EV market, different types of EVs should be considered in the load model. Using one specific type of EVs as the representative to model the load characteristics will bring large deviation. This chapter will expound what has been mentioned above in the DC side. The effect of the charging efficiency and power factor from AC side will be discussed in the next chapter.

A. The Numerical Process of a Single Charging Electric Vehicle Load

The numerical process of a charging EV load includes CC and CV two periods. In the CC period, the time-varying charge power based on the initial SOC and a certain charge current can be obtained from matching the battery voltage and the SOC with a polynomial. In the relatively short CV period, matching the charge power and the charge time directly is acceptable.

In the CC charge period, the following equation is satisfied

$$SOC = SOC_{initial} + I_c \Delta t / C \quad (2)$$

where Δt (h) is the charge period. The relationship between the battery voltage and the SOC based on a certain charge current $U_b(I_c, SOC)$ can be obtained by simulation or actual measurement. For example, when the charge current is 10 A, the relationship between the battery voltage and the SOC of the lithium-ion battery with rated capacity 92.8 Ah used before can be matched as

$$U_b(10, SOC) = -64SOC^4 + 183SOC^3 - 194SOC^2 + 97SOC + 354 \quad (3)$$

where the SOC is between 0.2 (that is 20%) and 0.9. The DC side charge power in the CC charge period satisfies the following equation

$$P_{CC}(SOC_{initial}, I_c, \Delta t) = IU_b(I_c, SOC) \quad (4)$$

When charge current is 10A, the DC side charge power can be obtained using (2), (3). For example, using (2), (3), the CC period charge power of the lithium-ion battery used before with 10A charge current and 0.2 initial SOC is expressed as (5).

$$P_{CC} = -0.087\Delta t^4 + 1.64\Delta t^3 - 11.60\Delta t^2 + 42.27\Delta t + 3674 \quad (5)$$

When the SOC rises to 0.9, the CV charge period of the lithium-ion battery used before begins with 375.8 V charge voltage. The matched polynomial of DC side charge power is

$$P_{CV}(t) = -0.00029t^4 + 0.048t^3 - 1.7t^2 - 71.4t + 3811 \quad (6)$$

where P_{CV} (W) is the charge power and t (min) is from 0 to 82. Equation (5) and (6) form the whole load characteristics of the used lithium-ion battery in the DC side, with the initial SOC 0.2 and the 10 A CC and 375.8 V CV charging. The power demand in the DC side and the related SOC profiles of this process are shown in Fig. 3.

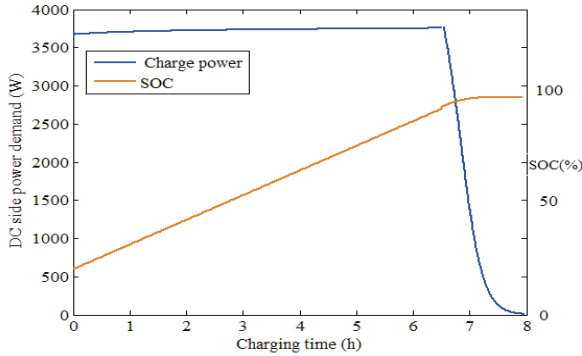


Fig. 3. 10 A CC and 375.8 V CV charging profile of the lithium-ion battery

The AC side EV load of this charge process is expressed as

$$P_{EV} = \begin{cases} \frac{-0.087\Delta t^4 + 1.64\Delta t^3 - 11.60\Delta t^2 + 42.27\Delta t + 3674}{\eta} \\ \frac{-0.00029t^4 + 0.048t^3 - 1.7t^2 - 71.4t + 3811}{\eta} \end{cases} \quad (7)$$

$$Q_{EV} = P_{EV} \tan \phi \quad (8)$$

where η and $\cos \phi$ (or λ) are the charging efficiency and the charging power factor of the EV charger, which will be expounded in detail in the subsequent chapter. The EV charge power used in power flow calculation of numerical simulation can be obtained by discretizing the power equations, like (7) and (8), in a certain time interval (e.g. every 5 min).

B. Electric Vehicle Loads and Their Changes Considering the Quantity and the Initial SOC Distribution

This section expounds the methodology for calculating the total charge power of a certain type of EVs in one charging period, considering the quantity and the initial SOC distribution. The quantity of the charging EV at one period, which has a certain distribution and many statistical data, can be estimated properly based on local EV sales, traffic and power prices [9]. The distribution of the initial SOC is mainly affected by users' driving habits, battery capacities, range anxiety, and so on. It is reasonable to take the distribution of the initial SOC as a normal distribution [9].

The "empirical law" of a normal distribution tells us that the probabilities of the variables, which are separately within

one, two or three times standard derivation from the mean value, are 68%, 95% and 99.7%, respectively. Considering the "empirical law" and the practical charge application, take the standard derivation as 0.1 is reasonable most of the time. The mean value of the initial SOC in a charging period can be estimated combining driving habits, battery capacities and the energy consumption of every mile. Based on a certain initial SOC distribution, the total charging EVs can be divided into several groups with a certain SOC interval (such as 0.05). The charging power change process of every group is different due to different initial SOC. If the case is ignored that charging EVs get off the charger halfway or new EVs start being charged in the considered charging period (in fact, this case can be reflected using one factor), the power change process until the end of charging of each group can be obtained using the numerical process described in the former section. Multiplying the power change process by the quantity of EVs of each group and then adding up the charging power of different groups at every time point, the charging power change process of the specific type of EVs can be estimated. Take the lithium-ion battery with rated capacity 92.8 Ah and charged with 10A CC and 375.8V CV as an example. Assume that at a point, 500 EVs with this type of battery are charging, and the initial SOC is a normal distribution with a mean 0.4 and a standard deviation 0.1. The estimated DC side charging load $P_{dc}(t)$ profile of this 500 EVs are shown in Fig. 4.

C. The Composite Load Model of Different Types of Electric Vehicles

Based on the current EV market status and the development trends, four types of EV models, as shown in table I, have been introduced as the representatives of the whole EV loads. In these four types, the fourth one is a low-speed EV model and electric buses are not included since their charge process is pretty regular and controllable. The load model of each type and the composite load model are calculated using the methodology described in the former two sections. Based on the practical applications, these four types of EVs are charged with 10 A, 10 A, 20 A, 25 A currents separately in the CC charge periods and the end voltages of the CC charge periods are used in the CV charge periods. Similar equations like (3) and (6) can be obtained for each type of EV following the procedure expounded in III.A. For example, the equations of the first type of EVs are like (9) and (10) where $0 \leq t \leq 52$ min.

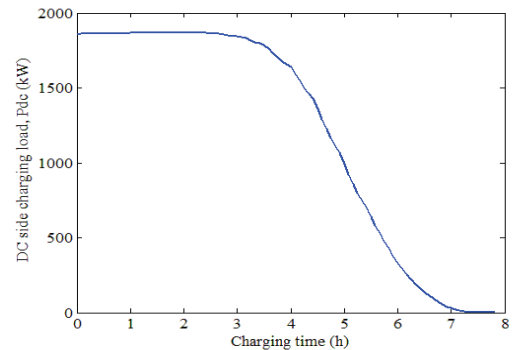


Fig. 4. Load profile of 500 EVs with a normal distribution initial SOC

TABLE I. FOUR TYPES OF MODELS SPECIFICATIONS

NO.	Normal voltage (V)	Capacity (Ah)	Maximum range (km)	Maximum charging period (h)
1	325	73.8	150-200	5-6
2	345.6	92.8	200-250	8-9
3	360	222.2	About 400	About 10
4	83	241	0-150	6-8

$$U_{bl}(10, SOC) = -58SOC^4 + 169SOC^3 - 180SOC^2 + 91SOC + 334 \quad (9)$$

$$P_{CV1}(t) = -0.00072t^4 + 0.108t^3 - 4.28t^2 - 35.2t + 3532 \quad (10)$$

Assume that the EVs considered in this study are included in one certain medium sized city, the daily mean distance driven for every EV is about 35 km and the energy consumption of every 100 km is 14-18 kWh. The quantity of each type of EVs is determined by the market sales. For the first type of EVs the mean value of the initial SOC is 0.33, which includes self-discharge and other energy consumption factors, and the quantity is 200. In the same way, the mean value of initial SOC and the quantities of the other three types are 0.33, 0.4, 0.45, and 400, 250, 400, respectively. The standard derivations are all assumed to be 0.1. The load model of each type and the composite load model in the DC side are shown in Fig. 5 and Fig. 6. From Fig. 5, the change profiles of these charge loads are similar, and the main differences of these four types EV loads are determined by the charge power and the charge period. Based on the work of this section, the load model of plug-in EVs in the AC side can be obtained by further consideration of the charging efficiency and power factor.

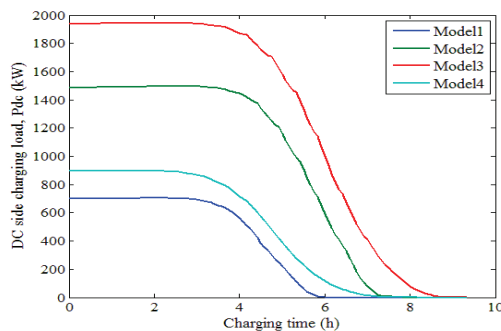


Fig. 5. Load models of four types EVs in the DC side

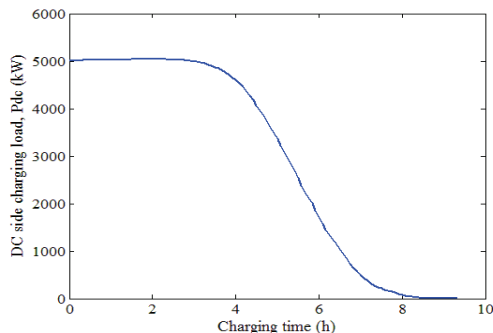


Fig. 6. The composite load model in the DC side

IV. MODELING OF ELECTRIC VEHICLE LOADS IN THE ALTERNATING CURRENT SIDE

According to Chinese electric vehicle industry standards, the on-board battery charger should work normally when the input voltage fluctuates within $\pm 15\%$ of the rated value (220 V) and the frequency is within $\pm 2\%$ of the rated value (50 Hz). In normal operating condition, the charging efficiency should be no less than 90%, and the power factor should be greater than 0.92. In practice, the well-designed on-board chargers keep the charging efficiency and power factor greater than 94% and 0.99 separately.

Using the on-board charger structure described in Section II, simulations of the CC charge period of the third type EVs have been performed, with the initial SOC 0.3. The changes of the efficiency and the power factor versus the AC frequency are shown in table II. It can be seen from table II that the changes of the efficiency and the power factor against the AC frequency are inside 0.01%, which is small enough to be ignored. The efficiency and the power factor versus the charging voltage in the CC period are shown in Fig. 7. These changes should be considered in the EV load model. The matched equations for the relationships shown in Fig. 7 are in the following where the unit for U_{ac} is 100 V

$$\eta(U_{ac}) = 0.1159U_{ac}^2 - 0.3039U_{ac} + 96.57 \quad (11)$$

$$\cos\varphi(U_{ac}) = -0.0126U_{ac}^4 + 0.125U_{ac}^3 - 0.468U_{ac}^2 + 0.785U_{ac} + 0.5 \quad (12)$$

The changes of the efficiency and the power factor in other SOC level and the CV period can also be obtained in the same way. After the calculation of $P_{dc}(t)$, the charging load in the DC side, following the procedures expounded in Section III, considering (11) and (12), the charging load of EVs in the AC side can be obtained as

$$P_{EV}(U_{ac}, t) = \frac{P_{dc}(t)}{\eta(U_{ac}(t))} \quad (13)$$

$$Q_{EV}(U_{ac}, t) = P_{EV}(U_{ac}, t) \tan\varphi(U_{ac}(t)) \quad (14)$$

Finally the numerical equations (13) and (14) can be used in quasi-steady state calculation and analysis of power systems with plug-in EVs.

TABLE II. CHARGING EFFICIENCY AND POWER FACTOR VERSUS FREQUENCY

NO.	Frequency (Hz)	Efficiency (%)	Power factor
1	48	96.45	0.9993
2	49	96.45	0.9993
3	49.5	96.46	0.9992
4	50	96.46	0.9992
5	50.5	96.46	0.9991
6	51	96.46	0.9991
7	52	96.47	0.9991

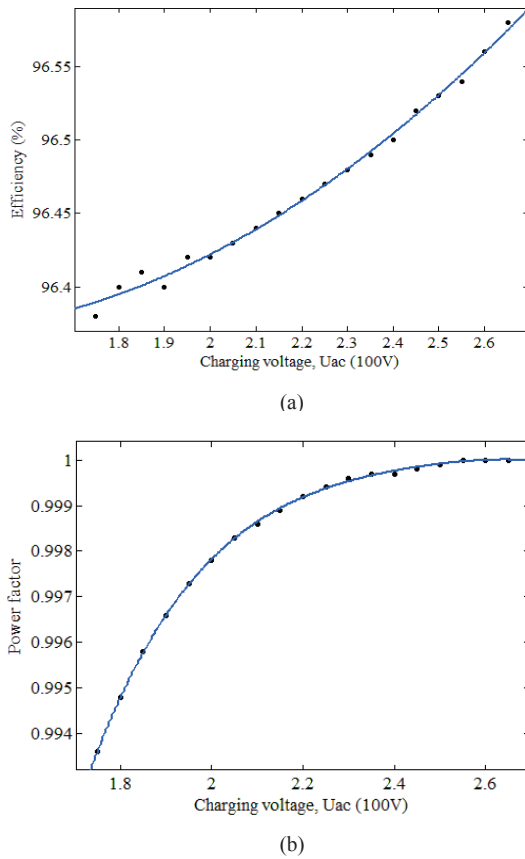


Fig. 7. Efficiency (a) and power factor (b) versus the charging voltage

V. CONCLUSION

This paper establishes the EV load model that can be applied to power system quasi-steady state analysis. A numerical load modeling methodology is introduced from two aspects, the DC side and the AC side of charging EVs. For the DC side load modeling, two charging processes, the CC and the CV periods, are considered. In addition, the quantity of charging EVs, the initial SOC distribution and the load change process during charging are included. Four types of the EV charging load model have been calculated using the proposed methodology. For the AC side load modeling, the changes of the charging efficiency and power factor versus power system frequency and voltage are analyzed. The EV load model applica-

ble to power flow calculation and other quasi-steady state analysis can be acquired by combining numerical modeling processes of the DC and the AC side.

The modeling method proposed in this paper can be applied to different types of EVs. There may be some differences in specific data, such as the quantity of the charging EVs and the initial SOC distribution due to different regions and habits. In addition, the effect of the BMS and the battery aging on the charging load can be studied further.

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